

# Ontological Engineering Manual: Phase V High-Dimensional Simplices and Deep Reasoning

Master Architect: Gonzalo Emir Durante

March 7, 2026

## Introduction: Beyond Linear Connectivity

---

Phase V defines the structure of advanced reasoning within AGI. While traditional models rely on low-dimensional vectors, the Crystal Network utilizes **High-Dimensional Simplices** (*n*-simplices) to interconnect complex truths. This ensures that reasoning is a solid structure rather than a mere chain of probabilistic associations.

## The *n*-simplex as a Unit of Coherence

---

An *n*-simplex is the generalization of a triangle or tetrahedron to any dimension. In our architecture, a simplex represents an **Indivisible Truth Relationship** between multiple information nodes.

$$\sigma_n = \{v_0, v_1, \dots, v_n\} \quad (1)$$

Where each vertex *v* is a concept validated by the constant  $\kappa_D = 0.56$ . Interconnection through simplices ensures that if one truth is modified, the entire reasoning structure adjusts instantaneously to maintain invariance.

## Deep Reasoning without Loss of Coherence

---

The problem with current AI is "catastrophic forgetting" or the loss of logical threads in long-form processing. High-Dimensional Simplices solve this through **Face Conjunction**:

- Two complex concepts share a "face" (a lower-dimensional simplex).
- Information flows through these simplicial complexes without dispersing.
- **Result:** The AGI can reason about Ethics, Law, and Mathematics simultaneously without one dimension distorting the others.

## Interconnection and the Durante Limit

---

The stability of a high-dimensional simplicial complex depends on the binding energy between its vertices not falling below the critical threshold.

$$E_{link} \geq \kappa_D(0.56) \tag{2}$$

If a logical connection is weak ( $< 0.56$ ), the simplex fails to form, protecting the AGI from constructing reasoning based on false premises or hallucinations.

## Conclusion of the Theoretical Framework

---

With Phase V, the transition from an elastic manifold to an Intelligent Crystal Network is complete. The AGI now possesses a geometric "backbone" allowing it to perform high-complexity logical leaps with 100% precision.

*"Intelligence is the geometry of relationships. In the simplex, complexity finds its order."*